

AI BACKLOG KILLER

From accumulated work to a controlled, measurable closure system

COO EXECUTIVE BRIEF

Scope: Inpatient Admission / Referral / Medical Records / Thai Utilization Management

Prepared for management discussion

Executive Summary

The Problem

- Three recurring backlog factories: Medical Record Requests, Procedure Cost Estimation, SIMB Cost Breakdown.
- The visible backlog mixes work-ready cases with cases waiting for data, decisions or approvals.
- Manual searching, rework and unclear ownership create hidden cycle time.

The Strategy

- Create one Backlog Control Tower with standard Intake → Classify → Prepare → Verify → Close.
- Use AI for extraction, triage, drafting, data preparation and follow-up.
- Keep human judgment for authorization, pricing exceptions, compliance and final release.

Expected Outcome

- More work becomes 'ready to work'.
- Managers see true backlog health—not just a large queue.
- AI reduces repetitive preparation while humans focus on high-risk decisions.
- 30-day target: establish control first, then scale automation.

Backlog Diagnosis

Three backlog factories require different AI patterns

Management implication: do not treat all backlog as one queue. Build a common control layer, then optimize each workflow separately.			
Backlog	Typical work	AI leverage	Human judgment
Medical Record Request	Request validation, document retrieval, completeness, release	High	High — authorization, privacy, final release
Procedure Cost Estimation	Procedure interpretation, component costing, comparable cases, quotation	High	High — clinical/price exceptions
SIMB Cost Breakdown	Extract charges, normalize, map categories, OR/Ward split, reconcile	Very High	Medium-High — mapping & final approval

The Core Diagnosis: Backlog Is a Flow Problem

Symptoms

- Old cases remain visible alongside new cases.
- Staff spend time rediscovering information.
- Cases stop at handoffs and approvals.
- Rework appears late in the process.

Likely Root Causes

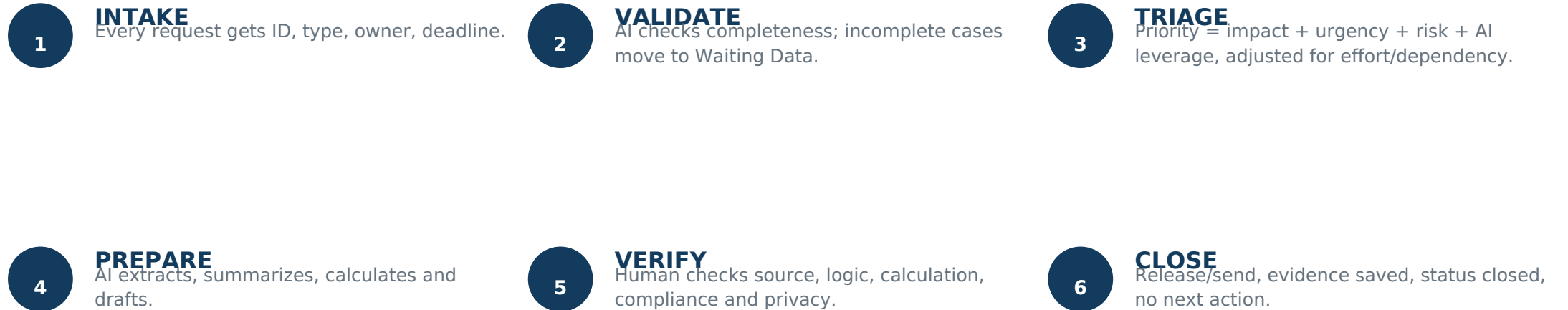
- No enforced Definition of Ready.
- No single source of truth for status/owner.
- Knowledge and historical examples are scattered.
- Waiting states are mixed with active work.

Design Principle

- AI prepares; human verifies.
- System records every status change.
- Only exceptions consume senior attention.
- Every case has a next action and deadline.

Root-cause hypothesis: a case-by-case operating model causes avoidable waiting and rework. This should be validated against actual backlog timestamps, owners and blockers.

Backlog Control Tower



Priority Matrix & Quick Wins

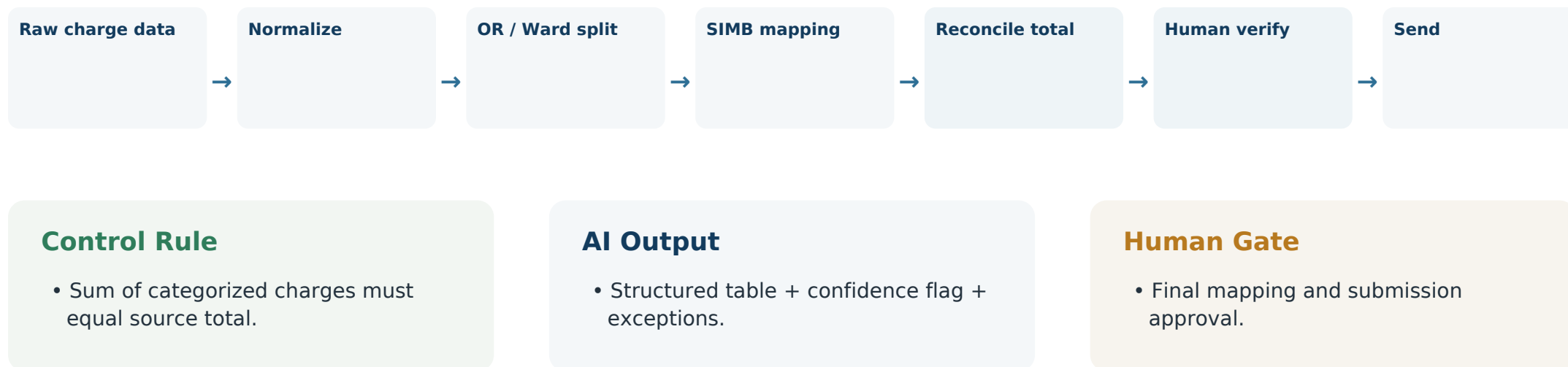
Priority	Action	Why now	Disposition
P1	SIMB Cost Breakdown pilot	Structured, repeatable, measurable; very high AI leverage	AUTOMATE / AI + HUMAN VERIFY
P1	Backlog Census + status cleanup	Creates one source of truth and exposes true work-ready volume	DO FIRST
P1	Separate Waiting Data / Decision / Approval	Removes false backlog and makes blockers visible	STANDARDIZE
P2	Procedure Cost Estimation knowledge base	Reduces repeated research and improves consistency	BUILD + AI ASSIST
P2	Medical Record Request intake/checklist	Reduces incomplete requests and rework	STANDARDIZE + AI ASSIST

AI Opportunity Map

Workflow step	AI can do	Human must do
Intake	Extract request fields; classify work type; identify missing data	Confirm exceptions and sensitive requirements
Document review	Summarize; compare request vs available documents; create checklist	Authorization / release decision
Price estimation	Extract procedure; find comparable cases; build component draft	Approve assumptions, exceptions and final price
SIMB	Clean data; map categories; separate OR/Ward; reconcile totals	Validate mapping and final submission
Management	Daily summary; aging analysis; follow-up list; draft escalation	Set priorities, resolve blockers, approve escalation

SIMB: Recommended First AI Pilot

Target flow



Medical Record Request: Risk-Controlled AI

AI Preparation

- Read request and extract required fields.
- Check requested date range and document types.
- Compare requested items with retrieved items.
- Generate missing-information list.
- Draft response / follow-up email.

Human Gate

- Verify requester authorization.
- Confirm consent / release conditions.
- Review sensitive or restricted content.
- Approve final release.
- Confirm evidence of transmission.

Do Not Automate Blindly

- Final disclosure decision.
- Interpretation of ambiguous authorization.
- Unapproved handling of identifiable health information.
- Any action outside hospital policy.

Management rule: use AI to reduce preparation time while keeping the release decision under accountable human control.

Verification System

Check	Question	Status logic
Three output states: ☐ Verified ☒ Need Review ☐ Cannot Confirm. The AI output is never the final evidence of correctness.		
Accuracy	Does output match the source?	☐ Verified / ☒ Review
Completeness	Are required fields present?	☐ Cannot Confirm if missing
Source & Date	Is the source identifiable and current?	☒ Review if stale/unknown
Calculation	Do totals reconcile?	☐ Cannot Close if mismatch
Logic & Contradiction	Are assumptions consistent?	☒ Review exceptions
Compliance & Privacy	Is use permitted under policy?	Human approval required

Definition of Done

Close criterion	Required evidence
Important: a case is not closed because “someone worked on it.” It is closed only when the requested outcome and evidence are complete.	
Request completed	Requested work delivered
Source verified	Source / version identified
Human review completed	Reviewer identified for high-risk items
Approval completed	Only where policy requires
Released / sent	Transmission evidence retained
Record updated	Status, closure date and owner recorded
No next action	No unresolved task remains

KPI: From Activity to Flow Performance

KPI	Baseline	30-day management target
Total Backlog	TBD — measure Day 1	↓ 50% or agreed target
Average Backlog Age	TBD	↓ materially
Cycle Time	TBD	↓ 20-30% pilot target
Human Hours / Case	TBD	↓ 20-30% on AI-pilot work
Rework / Error Rate	TBD	↓ without reducing control quality
Closed Cases / Month	TBD	↑ vs baseline
Overdue Rate	TBD	↓ toward zero for controlled queues

30-Day Action Plan

Timing	Management action	Deliverable
Day 1-2	Backlog census across 3 factories	Single master backlog + owner/status/deadline
Day 3-5	Clean statuses and separate waiting lanes	Ready / Working / Waiting / Escalated
Week 2	Standardize intake and Definition of Ready/Done	Checklists + templates
Week 3	Pilot AI on SIMB + one drafting workflow	Measured time / error / rework
Week 4	Review results and scale only what works	Automation roadmap + governance

Backlog Prevention System

Early Warning

- New work > closure capacity.
- Waiting cases age beyond threshold.
- No action recorded for X days.
- Backlog aging shifts upward.
- Rework rate increases.

Control

- Dashboard: New / Closed / Ready / Waiting / Overdue / Oldest.
- Auto reminders.
- Named owner for every case.
- Escalation threshold by work type.
- Weekly 15-minute review.

Governance

- Monthly process review.
- Version-controlled SOP and price/mapping knowledge.
- Approved AI-use boundaries.
- Audit trail for high-risk outputs.
- Periodic accuracy sampling.

Leading indicator: backlog is returning when new work enters faster than the team can reach “Closed,” or when waiting/rework starts increasing.

COO Decision Brief

Approve a 30-day Backlog Killer pilot

- 1. Establish one backlog inventory and common status model.
- 2. Pilot AI on SIMB Cost Breakdown and one document/drafting workflow.
- 3. Measure cycle time, human hours, error/rework and closure rate.
- 4. Define governance for privacy, human review and source verification.

START TODAY

First management question:

“How many open cases are truly ready to work today—and how many are actually waiting?”

Sources & Governance References

WHO — Ethics and governance of artificial intelligence for health

World Health Organization. 2021. Principles include protecting autonomy, promoting safety, transparency, accountability, equity and sustainable AI governance.

<https://www.who.int/publications/i/item/9789240029200>

WHO — Guidance on ethics & governance of AI for health

World Health Organization. Guidance supporting responsible use of AI in health, including human oversight and governance considerations.

<https://www.who.int/publications/i/item/9789240029200>

NIST AI Risk Management Framework

National Institute of Standards and Technology. AI RMF emphasizes managing AI risks through governance, mapping, measurement and management.

<https://www.nist.gov/itl/ai-risk-management-framework>

Note: operational targets and backlog scores in this presentation are proposed starting assumptions and should be recalibrated using actual hospital data.